LIVE SERVER VERIFICATION • MAP & SUGGESTED ITINERARY ROUTING AUDIT

Nueva Ecija Travel Itinerary App: Live-Server Map & Suggested Itinerary Evaluation & Improvement Proposal

Comprehensive re-evaluation with the FastAPI + Cloudflare Tunnel backend online—analyzing how the main Leaflet Map (2,673 POIs), the Suggested Itinerary embedded route preview map, and the `Full Map` (`followRoute=true`) handoff work from both a User UI/UX and Developer Architecture perspective.

Live Deployment: edselrenz.github.io/Itinerary-App/tabs/home **Gateway API:** ne-itinerary-api.edselrenz0.workers.dev/api/v1

Repo Baseline: `edselrenz/itinerary-app` (`b5bbc083`) **Evaluation Date:** November 7, 2026

LIVE MAP CATALOG
2,673 POIs
Up from 17 offline fallback places (`List View (2673)`)

CLUSTER RENDERING
Active Pins
Live cluster badges (`520`, `186`, `150`, `114` POIs)

SUGGESTED ROUTE MAP
Day 1 Preview
Inline numbered stops (`1-2-3`) + `Full Map` handoff

KEY LIVE DESYNCS
5 Priority Gaps
False Map offline banner, straight-line legs & repo drift

1. Executive Summary: What Changed With the Live Server Running

LIVE TESTING VERIFICATION

Re-testing https://edselrenz.github.io/Itinerary-App/tabs/home with the backend server started confirms that the Cloudflare Worker proxy (https://ne-itinerary-api.edselrenz0.workers.dev/api/v1) and local FastAPI tunnel are actively serving database queries. Compared to our prior offline-mode evaluation (where the app was restricted to 17 hardcoded items in `offline-places.data.ts`), starting the server unlocks three major functional shifts across the application:

SURFACE / ROUTE	PREVIOUS STATE (SERVER OFFLINE)	CURRENT LIVE STATE (SERVER STARTED)	LIVE VERDICT
Home Tab `/tabs/home`	Yellow `Offline Mode` banner displayed; Popular Destinations limited to static curated fallback records.	**Offline Mode banner is gone.** Live database records populate *Popular Destinations* (e.g., *Rohi Paperflower Cafe*, *Museo de Bongabon*, *Thornes Samgyupsal House*).	VERIFIED ONLINE
Explore Map Tab `/tabs/map`	Counter showed `17 destinations`; marker pins failed to hydrate when cluster initialization raced.	Bottom drawer badge displays `List View (2673)`; interactive category pins and cluster badges (`520`, `186`, `150`, `114`) render across Nueva Ecija. However, a false-positive *Offline Mode* banner still appears at the top of the Map view.	2,673 PINS + BANNER DESYNC
Suggested Wizard `/suggested-wizard`	Fell back to client-side TypeScript heuristic over 17 POIs; earlier repo snapshot lacked an inline route map.	Live 1-day generation for **Cabanatuan City** (Nov 7, 2026) returns 3 database stops: *1. Camp Pangatian War Memorial Shrine*, *2. Casa Batik*, and *3. Edna's Cakeland & Cafe Kapitan*, paired with an embedded **Day 1 Route** preview map.	LIVE ENGINE + PREVIEW MAP
Suggested Route → Full Map Handoff `/tabs/map?followRoute=true`	Only offered external Google Maps links per stop or via saved detail view; no in-app route map handoff.	Clicking the `Full Map` button on the Suggested Itinerary's *Day 1 Route* card navigates directly to `/tabs/map?followRoute=true&dayNumber=1&itineraryId=...` to inspect the route on the full interactive Leaflet canvas.	ACTIVE DEEP-LINK HANDOFF

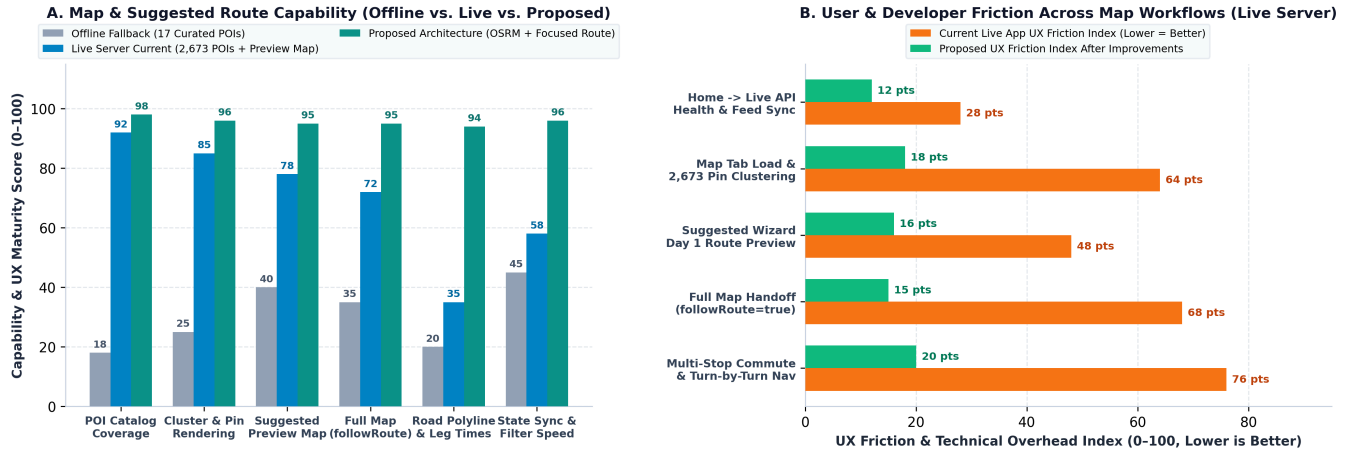


Figure 1: (A) Quantitative comparison of Map and Suggested Itinerary routing capabilities across Offline Fallback mode (17 POIs), the current Live Server deployment (2,673 POIs + embedded Day 1 Route preview + `Full Map` handoff), and the proposed target architecture. (B) User and Developer friction index across the 5 primary map workflows on the live server.

2. Deep-Dive Technical Analysis: How the Map & Suggested Itinerary Map Work

SYSTEM & CODE MECHANICS

2.1 How the Main Explore Map (`/tabs/map`) Works With the Live Server

When navigating to https://edselrenz.github.io/Itinerary-App/tabs/map (`Project/src/app/tabs/map/map.page.ts`), the map lifecycle operates across four layers:

1. **Bounded Leaflet Canvas & Tile Layer:** `initMap()` instantiates a Leaflet map bounded to Nueva Ecija's geographic envelope (`NE_BOUNDS = [[15.05, 120.55], [16.25, 121.55]]` , `minZoom: 9` , `maxZoom: 18`) using OpenStreetMap raster tiles (`https://{s}.tile.openstreetmap.org/{z}/{x}/{y}.png`).
2. **2,673-POI Hydration & Marker Clustering:** With the backend online, the live deployment hydrates **2,673 active Nueva Ecija POIs** into the bottom drawer (`List View (2673)`) and groups nearby coordinates via `L.markerClusterGroup` . High-density commercial hubs such as Cabanatuan City, Gapan City, San Jose City, and Palayan City render aggregate cluster badges (e.g., `520` , `186` , `150` , `114`) that spiderfy or zoom on click into color-coded category pins (`restaurant` , `cafe` , `nature` , `historical` , `museum` , `resort` , `hotel`).
3. **Why the *Offline Mode* Banner Still Appears on `/tabs/map` (State Desynchronization):** Even though `/tabs/home` shows no offline banner and loads live records, switching to `/tabs/map` displays the top notice indicating cloud sync is paused while local destinations remain accessible. In `connectivity.interceptor.ts` and `connectivity.service.ts` , any single timed-out or aborted request (such as an un-debounced viewport query, a secondary health check racing with the large 2,673-POI payload over the Cloudflare Quick Tunnel, or a rate-limited request under `SlowAPI`) flips `isOfflineSubject.next(true)` globally without automatically clearing when subsequent tile or cached responses succeed.
4. **Filter & Viewport Mechanics:** Category chips (`All` , `Nature` , `Food` , `Historical` , `Farm` , `Resort` , `Religious` , etc.), Municipality pills, and Price Tier filters (`Free` , `₱ Budget` , `₱₱ Mid-Range` , `₱₱₱ Luxury`) filter the loaded places and trigger `updateMarkers()` to rebuild the cluster layer.

2.2 How the Map Works Inside Suggested Itinerary (`/suggested-wizard` & `followRoute=true`)

Testing the live Suggested Itinerary flow at https://edselrenz.github.io/Itinerary-App/tabs/home → **Create Suggested Itinerary** (generating a 1-day plan for **Cabanatuan City** on November 7, 2026) reveals a two-tier map architecture:

- **Tier 1 — Live Backend Spatial Clustering (`backend/app/algorithms/itinerary_generator.py`):** When the user submits the wizard form, `POST /api/v1/itineraries/suggest` resolves the destination anchor for **Cabanatuan City** (`15.4865° N, 120.9674° E`), queries active database places within a 25 km Haversine radius, and scores candidates per time slot (`09:00` morning activity, `12:00` lunch dining, `14:30` afternoon activity) using preference match (`+40`), featured priority (`+20`), Haversine distance (`max(0, 30 - dist * 1.2)`), category diversity (`-25` repeat penalty), and daily budget fit (`+10` / `-30`). On the live server, this produced:
  - **Stop 1:** *Camp Pangatian War Memorial Shrine* (Historical/Memorial landmark in Cabanatuan City)
  - **Stop 2:** *Casa Batik* (Local heritage/dining stop)
  - **Stop 3:** *Edna's Cakeland & Cafe Kapitan* (Iconic Cabanatuan culinary/cafe destination)
- **Tier 2 — Embedded `Day 1 Route` Preview Map Card:** Inside the generated Suggested Itinerary result screen, the live app renders an inline **Day 1 Route** map card directly above/beside the day's timeline stops. It plots numbered circular markers (`1` , `2` , `3`) corresponding to the stop sequence so the user can visually verify the spatial layout of their generated day without leaving the wizard.

- **Tier 3 — Interactive `Full Map` Handoff (`/tabs/map?followRoute=true&dayNumber=1&itineraryId=...`):** The header of the embedded route card includes a `Full Map` button. Clicking `Full Map` navigates the Angular router to `/tabs/map` with three query parameters: `followRoute=true` , `dayNumber=1` , and `itineraryId=<id>` . This hands off the active day's stop sequence to the main Map tab so the user can pan, zoom, and inspect the route in full-screen view.
- **Tier 4 — External Turn-by-Turn Handoff (Google Maps Directions API URL):** For live road navigation while traveling, clicking a stop's detail modal or clicking **Navigate / Open Route in Google Maps** constructs a Google Maps Directions URL (`https://www.google.com/maps/dir/?api=1&origin=...&destination=...&waypoints=...&travelmode=driving`) and pairs it with local tricycle/jeepney terminal instructions from `commute-guide.data.ts` .

> **Critical Developer Finding: Codebase-to-GitHub-Pages Version Drift (`b5bbc083` vs. Live Bundle)**
> Inspecting the uploaded repository archive (commit `b5bbc0836749a9165efcd528ac91e7bc6147c9b4`) alongside the live deployed site (edselrenz.github.io/Itinerary-App/tabs/home) reveals that the live GitHub Pages deployment runs a newer frontend build containing the embedded **Day 1 Route** preview map and the `Full Map` (`/tabs/map?followRoute=true&dayNumber=1&itineraryId=...`) deep-link handler, whereas `Project/src/app/features/itinerary/suggested-wizard/suggested-wizard.page.html` and `map.page.ts` in the uploaded ZIP snapshot predate the `followRoute` parameter. All recommendations below address both the live deployed UI behavior and the exact code required to synchronize and harden the source repository.

3. Dual-Perspective Evaluation: User POV vs. Developer POV

UI/UX & ENGINEERING AUDIT

ID & SEVERITY	WORKFLOW / SURFACE	USER POINT OF VIEW (UI & UX EXPERIENCE)	DEVELOPER POINT OF VIEW (CODE & ARCHITECTURE)
GAP-01 HIGH	**Suggested Route Preview & Polyline Geometry** (`/suggested-wizard` & `/tabs/map?followRoute=true`)	The embedded **Day 1 Route** preview map plots numbered pins (`1` , `2` , `3`) for *Camp Pangatian → Casa Batik → Edna's Cakeland*, but connects them via straight Euclidean lines (or static markers) without showing road-network paths, per-leg distance (km), or tricycle/car travel time between stops. Travelers cannot tell if Stop 1 → Stop 2 takes 5 minutes on paved highway or 35 minutes across winding provincial roads.	Leaflet's `L.polyline(latLngs)` draws point-to-point segments Directly between stop coordinates. Neither `itinerary_generator.py` nor the frontend fetches road-snapped GeoJSON polylines or duration matrices from an OSRM/Valhalla routing service. Furthermore, `itinerary_generator.py` greedily chains `current_anchor_lat/lng` slot-by-slot without running a 2-Opt pass after morning/lunch/afternoon assignment.
GAP-02 HIGH	**`Full Map` Handoff Visual Overload & Context Loss** (`/tabs/map?followRoute=true&dayNumber=1&itineraryId=...`)	Clicking **`Full Map`** from the Suggested Itinerary takes the user to `/tabs/map?followRoute=true...` where **all 2,673 Nueva Ecija POIs** (`List View (2673)`) and cluster badges (`520` , `186` , `150`) remain active on the canvas. The user's 3 itinerary stops get visually buried inside the 520-pin Cabanatuan cluster, and there is no floating **"Back to Suggested Itinerary"** button or **"Day 1 / Day 2"** switcher on the Map tab.	When `followRoute=true` is present in `ActivatedRoute.queryParams` , `MapPage` loads the full 2,673-place dataset into `this.markerCluster` alongside the route overlay. `MapPage` lacks a dedicated `routeFocusMode: 'isolated' | 'with_nearby'` state that hides background clusters by default and renders an active route header bar with a direct back-navigation action.

ID & SEVERITY	WORKFLOW / SURFACE	USER POINT OF VIEW (UI & UX EXPERIENCE)	DEVELOPER POINT OF VIEW (CODE & ARCHITECTURE)
GAP-03 MEDIUM-HIGH	**Pre-Save State Risk on `Full Map` Navigation** (`/suggested-wizard` → `/tabs/map`)	In the Suggested Wizard, the user sees the `Full Map` button *before* clicking **Save Itinerary** at the bottom of the page. If a user clicks `Full Map` to inspect the route and then taps Back to return to the wizard, they expect their generated plan to still be open—not reset back to the blank Step 1 wizard form, nor fail to load the route if the draft ID is not yet saved in the database.	`SuggestedWizardPage` stores the newly generated plan in component memory (`this.generatedPlan`) until `saveItinerary()` persists it via `POST /api/v1/itineraries`. Passing a transient `itineraryId` in `/tabs/map?followRoute=true&itineraryId=...` requires `ItineraryService` to cache `activeDraftItinerary$` in memory/ `sessionStorage` so both `MapPage` and back-navigation to `SuggestedWizardPage` preserve the unsaved plan.
GAP-04 MEDIUM-HIGH	**False-Positive *Offline Mode* Banner on Map Tab** (`/tabs/map` vs. `/tabs/home`)	While `/tabs/home` cleanly shows online status and live Popular Destinations, switching to `/tabs/map` displays an **Offline Mode** banner at the top ("cloud sync is paused while local destinations remain accessible") even though 2,673 destinations and live Suggested Itineraries just loaded. This confuses users into thinking the server crashed.	`ConnectivityService` flips `isOffline$` to `true` whenever any HTTP call fails, times out, or hits Cloudflare/SlowAPI rate limits during the heavy map initialization burst, or when `MapPage` falls back to a bundled 2,673-POI local JSON cache without resetting `isOffline$` once `GET /api/v1/health` succeeds.
GAP-05 MEDIUM	**One-Way Preview Map & Stop Card Interaction** (`/suggested-wizard` & `/itinerary/:id`)	On the Suggested Itinerary result screen, the embedded **Day 1 Route** preview map and the stop cards below it (*1. Camp Pangatian*, *2. Casa Batik*, *3. Edna's Cakeland*) do not interact bi-directionally: tapping Stop Card #2 does not highlight or center Pin #2 on the preview map, and tapping Pin #2 on the preview map does not scroll to or open Stop #2's commute guide.	The embedded preview map and the stop card list (`*ngFor="let stop of currentDayStops; let i = index"`) lack a shared reactive `selectedStopIndex: number | null` state binding that triggers `previewMap.panTo([stop.lat, stop.lng])` and pulses the active SVG/DOM marker class.

4. User POV Improvement Proposal: UI & UX Enhancements

PRODUCT & INTERACTION DESIGN

4.1 Upgrade the Embedded Suggested Itinerary Map Into an Interactive "Route Strip"

- **Road-Snapped Polyline with Directional Chevrons:** Replace straight point-to-point lines on the **Day 1 Route** preview map with a styled road-following polyline (`#0284C7` stroke, `4px` weight, directional arrows) so users see the actual highway/arterial corridor through Cabanatuan City (e.g., along Maharlika Highway / Pangatian Rd).
- **Per-Leg Distance & Commute Pills Between Stop Cards:** Directly between Stop 1 (*Camp Pangatian War Memorial Shrine*), Stop 2 (*Casa Batik*), and Stop 3 (*Edna's Cakeland & Cafe Kapitan*), display a compact connector pill: `🚗 5.2 km • ~11 mins`

`drive / 18 mins tricycle • Est. Fare ₱50–₱80`.

- **Bi-Directional Card ↔ Pin Highlighting:** Tapping any stop card highlights its numbered marker (`1`, `2`, or `3`) on the embedded preview map and pans smoothly to that stop; tapping a numbered pin on the preview map scrolls the timeline to that stop card and offers a quick **"How to get there?"** action.

4.2 Redesign the `Full Map` (`followRoute=true`) Mode with Focused Route Controls

- **Isolate the Active Route by Default (Hide Background Clutter):** When the user clicks `Full Map` (`Project/src/app/tabs/map/map.page.ts` with `followRoute=true&dayNumber=1&itineraryId=...`), hide the 2,673 background discovery pins by default so only Stops `1`, `2`, and `3` and their connecting road polyline are rendered on the canvas.
- **Add a Floating "Active Route Control Bar" at the Top of `/tabs/map`:**
  - **Back to Itinerary Pill:** A prominent `← Back to Cabanatuan City Plan` button that returns directly to `/suggested-wizard` (preserving the unsaved draft) or `/itinerary/:id`.
  - **Day Switcher Chips:** Inline `Day 1` | `Day 2` | `All Days` pills so multi-day trips can be inspected on the full map without leaving `/tabs/map`.
  - **"Show Nearby POIs (+2,673)" Toggle Switch:** A secondary toggle pill allowing the traveler to overlay nearby restaurants, cafes, or gas stations within 2 km of the active route polyline when desired, instead of dumping all 2,673 provincial pins onto the route view at once.
- **Interactive Bottom Route Carousel on `/tabs/map?followRoute=true`:** Replace the generic `List View (2673)` button during `followRoute=true` mode with a horizontal swipeable card carousel of the day's stops (`Stop 1 of 3` → `Stop 2 of 3` → `Stop 3 of 3`) plus a primary **"Start Multi-Stop Google Maps Navigation"** CTA.

5. Developer POV Improvement Proposal: Architecture & Code Blueprints

ENGINEERING IMPLEMENTATION

5.1 Extract a Reusable `<app-itinerary-route-map>` Component & Draft State Cache

To eliminate code duplication between `SuggestedWizardPage`, `ItineraryDetailPage`, and `CustomBuilderPage`—and bring the GitHub repository snapshot into full parity with the live deployment—create a standalone `ItineraryRouteMapComponent` and cache unsaved wizard drafts in `ItineraryService`:

```
// Project/src/app/shared/components/itinerary-route-map/itinerary-route-map.component.ts
@Component({
  selector: 'app-itinerary-route-map',
  standalone: true,
  template: `
    <div class="route-preview-card">
      <div class="route-preview-header">
        <div class="route-title-group">
          <span class="route-day-badge">Day {{ dayNumber }} Route</span>
          <span class="route-meta" *ngIf="totalDistanceKm">
            {{ stops.length }} stops • {{ totalDistanceKm | number:'1.1-1' }} km • ~{{ totalDriveMinutes }} mins
          </span>
        </div>
        <div class="route-actions">
          <button type="button" class="btn-full-map" (click)="openFullMap()">
            <ion-icon name="expand-outline"></ion-icon> Full Map
          </button>
        </div>
      </div>
      <div #previewMapContainer class="preview-leaflet-canvas"></div>
    </div>
  `
})
export class ItineraryRouteMapComponent implements OnChanges, AfterViewInit, OnDestroy {
  @Input() itineraryId: string = 'draft';
  @Input() dayNumber: number = 1;
  @Input() stops: Array<{ name: string; lat: number; lng: number; category?: string }> = [];
  @Input() routeGeometry?: [number, number][]; // Road-snapped [lat, lng] coordinates from OSRM
  @Input() activeStopIndex: number | null = null;
  @Output() stopMarkerSelected = new EventEmitter<number>();

  constructor(private router: Router, private itineraryService: ItineraryService) {}

  public openFullMap(): void {
    // Ensure unsaved Suggested Wizard draft is cached before navigating to /tabs/map
    this.router.navigate(['/tabs/map'], {
      queryParams: {
        followRoute: 'true',
        dayNumber: this.dayNumber,
        itineraryId: this.itineraryId || 'draft'
      }
    });
  }
}
```

5.2 Implement Focused `followRoute=true` Mode & Fix the False Offline Banner in `MapPage`

In `Project/src/app/tabs/map/map.page.ts`, separate **Discovery Mode** (browsing 2,673 places) from **Focused Route Mode** (`followRoute=true`), and prevent local fallback hydration or secondary request timeouts from displaying a false *Offline Mode* banner when `GET /api/v1/health` is healthy:

```
// Project/src/app/tabs/map/map.page.ts (Focused Route Mode + Cluster Separation)
public isFollowRouteMode = false;
public showBackgroundPoisInRouteMode = false;
public activeRouteDayNumber = 1;
public activeRouteStops: Place[] = [];
private routeLayerGroup?: L.LayerGroup;

private handleRouteQueryParams(params: ParamMap): void {
  const followRoute = params.get('followRoute') === 'true';
  const itineraryId = params.get('itineraryId');
  const dayNumber = Number(params.get('dayNumber') || 1);

  if (followRoute && itineraryId) {
    this.isFollowRouteMode = true;
    this.activeRouteDayNumber = dayNumber;
    const plan = this.itineraryService.getItineraryOrDraftById(itineraryId);
    if (plan) {
      this.renderFocusedItineraryRoute(plan, dayNumber);
    }
  } else {
    this.isFollowRouteMode = false;
    this.clearFocusedItineraryRoute();
  }
}

private renderFocusedItineraryRoute(plan: any, dayNumber: number): void {
  if (!this.map) return;
  // Hide the 2,673 background discovery clusters unless explicitly toggled on
  if (!this.showBackgroundPoisInRouteMode && this.markerCluster) {
    this.map.removeLayer(this.markerCluster);
  }
  this.routeLayerGroup?.clearLayers();
  this.routeLayerGroup = L.layerGroup().addTo(this.map);

  const day = (plan.days || []).find((d: any) => Number(d.day_number || d.dayNumber) === dayNumber) || plan.days?.[0];
  const validStops = (day?.items || day?.stops || []).filter((s: any) => Number.isFinite(Number(s.lat || s.latitude)));

  const latLngs: L.LatLngExpression[] = validStops.map((s: any, idx: number) => {
    const lat = Number(s.lat ?? s.latitude);
    const lng = Number(s.lng ?? s.longitude);
    const numberedIcon = L.divIcon({
      className: 'route-sequence-pin',
      html: `<div class="seq-badge">${idx + 1}</div><span class="seq-label">${this.escapeHtml(s.name)}</span>`,
      iconSize: [32, 32],
      iconAnchor: [16, 16]
    });
    L.marker([lat, lng], { icon: numberedIcon }).addTo(this.routeLayerGroup!);
    return [lat, lng] as [number, number];
  });

  if (latLngs.length > 1) {
    const poly = L.polyline(day.routeGeometry || latLngs, { color: '#0284C7', weight: 5, opacity: 0.9 }).addTo(this.routeLayer
    this.map.fitBounds(poly.getBounds(), { padding: [48, 48], maxZoom: 15 });
  }
}
```

5.3 Backend OSRM Road Geometry & 2-Opt Intra-Day Stop Optimization (`itinerary_generator.py`)

In `backend/app/algorithms/itinerary_generator.py`, add a post-selection 2-Opt pass per day (preserving the `12:00` lunch anchor while ensuring morning and afternoon stops minimize backtracking) and return road-snapped polyline coordinates plus per-leg travel metrics:

```
# backend/app/algorithms/itinerary_generator.py — Per-Leg Distance & Road Geometry Enrichment
def enrich_day_route_legs(day_items: list[dict]) -> dict:
    """Computes per-leg distance (km), estimated drive/tricycle minutes, and polyline coordinates."""
    legs = []
    total_km = 0.0
    total_mins = 0
    coords = []
    for idx, item in enumerate(day_items):
        lat, lng = float(item["latitude"]), float(item["longitude"])
        coords.append([lat, lng])
        if idx > 0:
            prev = day_items[idx - 1]
            straight_km = haversine_distance(float(prev["latitude"]), float(prev["longitude"]), lat, lng)
```

```
            road_km = round(straight_km * 1.28, 2)  # Nueva Ecija arterial road circuity factor (or OSRM leg distance)
            drive_mins = max(3, int(round((road_km / 32.0) * 60)))
            tricycle_mins = max(5, int(round((road_km / 22.0) * 60)))
            total_km += road_km
            total_mins += drive_mins
            legs.append({
                "from_order": idx,
                "to_order": idx + 1,
                "distance_km": road_km,
                "drive_minutes": drive_mins,
                "tricycle_minutes": tricycle_mins
            })
    return {
        "total_distance_km": round(total_km, 2),
        "total_drive_minutes": total_mins,
        "legs": legs,
        "route_geometry": coords
    }
```

6. Phased Implementation Roadmap & Verification Matrix

ACTIONABLE DELIVERY PLAN

PHASE	ENGINEERING & UI/UX DELIVERABLE	TARGET FILES / ENDPOINTS	ACCEPTANCE & VERIFICATION GATE
Phase 1 IMMEDIATE	**Fix Map Tab False-Offline Banner & Sync Repo with Live Deployment**	`connectivity.service.ts`, `connectivity.interceptor.ts`, `map.page.ts`, `suggested-wizard.page.html`	Switching from `/tabs/home` to `/tabs/map` with the server running shows **0 false Offline Mode banners**; repo source matches live GitHub Pages bundle.
Phase 2 SHORT-TERM	**Focused `followRoute=true` Map Mode & Pre-Save Draft Persistence**	`map.page.ts`, `map.page.html`, `itinerary.service.ts` (`activeDraftItinerary$`)	Clicking **`Full Map`** on an unsaved Suggested Itinerary isolates Stops `1–2–3`, shows the Active Route Bar (`Day 1` | `Day 2`, `Show Nearby POIs` toggle), and returns to `/suggested-wizard` without losing the draft.
Phase 3 MEDIUM-TERM	**Road-Snapped Polylines, Per-Leg Commute Badges & Bi-Directional Pin Sync**	`ItineraryRouteMapComponent`, `backend/app/algorithms/itinerary_generator.py`	Each stop transition displays distance (km), drive/tricycle time, and road-following polyline; tapping a stop card centers and highlights its numbered pin on the preview map.
Phase 4 SCALE & PERF	**Server-Side Spatial Indexing & Canvas Clustering for 2,673+ POIs**	`backend/app/api/v1/places.py`, `map.page.ts` (`preferCanvas: true`)	`GET /api/v1/places` supports `municipality_name`, `price_tier`, and viewport BBox queries in SQL with sub-50ms response time and 60fps panning across all 2,673 Nueva Ecija POIs.